Bogleheads

Value averaging

Value averaging uses a formula, shown below, to guide how much you invest into a given investment at a specific time. The emphasis is on establishing a portfolio target value or "value path". Value averaging aims to periodically increase the investment's value by this calculated amount.

Whenever a portfolio under-performs, you will have to make a larger investment to make up for the under-performance. The converse is also true, so that if the portfolio outperforms its targeted rate of return, then then you do not purchase more shares. You can think of value averaging as a combination of dollar cost averaging and portfolio rebalancing.[1]

Former Harvard professor Michael E. Edleson introduced value averaging in his book *Value Averaging*, first published by Wiley in 1988.[2]

How value averaging works

The key idea is to create a *Value Path* which will describe the investment's target level for each time period. You must simply make the proper investment or sale at each period so that the holdings are equal to the target value.

The variables and the formula for the Value Path are as follows.[3]

Variables:

- **t** = Time period (can be months, quarters, years, etc.)
- $\mathbf{V_t}$ = Target value of investment at time period **t**
- **C** = Target initial contribution per period
- **r** = Expected rate of growth per period of investment
- **g** = Expected rate of growth per period of contribution
- **R** = Average rate of growth of investment and contribution

To simplify the calculation, we will set **R** = (**r** + **g**) / 2. Thus, **R** is just an average of the growth rates of the investment and the contribution amount.

Now, the formula for the Value Path:

$$\mathbf{V_t} = \mathbf{C} \times \mathbf{t} \times (1 + \mathbf{R})^{\mathbf{t}}$$

Now, you can simply use this formula to generate a Value Path for your investment. There are two different approaches for this depending on your needs. If you have a specific goal in mind, for example you need $100,000 in 10 years for your son's college expenses, you can start with the final value of the Value Path and use that to determine the required value of your initial contribution (**C**). So, for our example, if we were working with monthly time periods, we would want $\mathbf{V_{120}}$ = 100,000. Then, we solve $\mathbf{V_{120}} = \mathbf{C} \times 120 \times (1 + \mathbf{R})^{\mathbf{120}}$.

On the other hand, if you have no specific savings goal but do know the amount of initial contribution that you are comfortable with, then you can plug in a value of **C** and solve for all the values of $\mathbf{V_t}$.

It is a good idea to use a spreadsheet to create your Value Paths and track your progress. An important thing to remember about the Value Path is that you should remain flexible and update your Value Path if circumstances change. For instance, if the rate of growth for your investment or your ability to contribute changes, you should re-calculate the Value Path from your current point.

The Value Path formula depends heavily on your estimates of the rate of growth of the investment asset. Because of this, it is very important to provide as realistic rates as possible.

Example

The following spreadsheet table compares value averaging with dollar cost averaging:

Dollar Cost vs Value Averaging : Dollar Cost vs Value Averaging

	Contribution	$1,000.00								
		Dollar Cost Averaging				Value Averaging				Amoun
Period	Market Price	Amount Invested During Period	Shares Purchased	Shares Owned	Value	Amount Invested During Period	Shares Purchased	Shares Owned	Value Path (Amount Required)*	Dollar C
1	$10.00	$1,000.00	100	100	$1,000.00	$1,000.00	100	100	$1,000.00	
2	$11.00	$1,000.00	91	191	$2,100.00	$900.00	82	182	$2,000.00	
3	$12.00	$1,000.00	83	274	$3,290.91	$818.18	68	250	$3,000.00	
4	$13.00	$1,000.00	77	351	$4,565.15	$750.00	58	308	$4,000.00	
5	$12.00	$1,000.00	83	434	$5,213.99	$1,307.69	109	417	$5,000.00	
6	$11.00	$1,000.00	91	525	$5,779.49	$1,416.67	129	545	$6,000.00	
7	$10.00	$1,000.00	100	625	$6,254.08	$1,545.45	155	700	$7,000.00	
8	$9.00	$1,000.00	111	737	$6,628.67	$1,700.00	189	889	$8,000.00	
9	$10.00	$1,000.00	100	837	$8,365.19	$111.11	11	900	$9,000.00	
10	$11.00	$1,000.00	91	927	$10,201.71	$100.00	9	909	$10,000.00	
11	$12.00	$1,000.00	83	1011	$12,129.14	$90.91	8	917	$11,000.00	
12	$11.00	$1,000.00	91	1102	$12,118.38	$1,916.67	174	1091	$12,000.00	
	Average:	$1,000.00	92			$971.39	91			

	Average Cost	Total Cost	Current Value	Current Gain	Return (IRR)
Dollar Cost Average	$10.89	$12,000.00	$12,118.38	$118.38	0.15%
Value Average	$10.69	$11,656.68	$12,000.00	$343.32	0.43%
Lump Sum	$10.00	$12,000.00	$13,200.00	$1,200.00	10.00%

Vary market price (Column C) to see effects on Value Averaging contributions (Column H).

Dollar Cost vs Value Averaging < >

(View Google Spreadsheet in browser (http://spreadsheets.google.com/ccc?key=1sqEPSLZUCSvNxVyO5D4Dg_gSAL56gfL7Jia4mBRxGmo), then File --> *Download as* to download the file.)
Note: If the spreadsheet is blank, select a different sheet, then back to that sheet. The image will be refreshed.

Cautions and qualifications

- Value averaging adds a growth factor (by formula, consisting of estimates of both the investment's return as well as growth in the contribution level) to the regular periodic investment of savings flows. If your contribution amount is not expected to grow, you should set this factor at 0% so that investment growth is the only growth variable.
- Value averaging may need both purchases and sales of the underlying investment, based on the investment performance of the investment. It requires a cash account for prospective purchases as well as any sales proceeds.
- Value averaging sales can realize taxable capital gains. You might want to restrict it to tax advantaged accounts; or alternately, adopt a policy constraint forbidding or delaying sales in the taxable account.
- You have to be careful in a declining market, as you will need to have additional funds available for investment. As time progresses and the account value grows, you may need even more funds to reach the amount required for the chosen period. If the market is volatile near the end of the investing time frame, you will need a lot of additional funds. This may catch investors unprepared for this additional savings requirement.
- Value averaging most often provides a lower average cost per share than does DCA, and also provides for a higher internal rate of return (IRR). This does not, however, mean that value averaging will result in a higher realized profit.[4]

See also

- Dollar cost averaging

References

1. Bruce Ramsey. "The Value Averaging Investment Strategy" (https://www.valueaveraging.ca/research/The%20Value%20Averaging%20Investment%20Strategy.pdf) (PDF). VA Investment Software. Retrieved August 13, 2023.
2. Michael E. Edleson (2006). *Value Averaging: The Safe and Easy Strategy for Higher Investment Returns*. Wiley. ISBN 978-0-470-04977-8.
3. "Value Averaging" (https://web.archive.org/web/20210508133532/https://www.sigmainvesting.com/advanced-investing-topics/value-averaging). Sigma Investing. 2007. Archived from the original (https://www.sigmainvesting.com/advanced-investing-topics/value-averaging) on May 8, 2021.

4. Simon Hayley (March 11, 2011). "Value Averaging and the Automated Bias of Performance Measures" (https://ssrn.com/abstract=1606347). Retrieved August 13, 2023.. Available at SSRN.

External links

- Choosing Between Dollar-Cost and Value Averaging (https://www.investopedia.com/articles/stocks/07/dcavsva.asp) - Investopedia
- DCA versus Value Averaging (https://www.financialwisdomforum.org/gummy-stuff/DCA-vs-VA.htm) - gummy stuff
- How Value Averaging adds Value (https://web.archive.org/web/20131206124712/https://valueaveraging.ca/docs/Study_How_Value_Averaging_adds_Value.pdf) - Bruce Ramsey (November 15, 2010)
- Bogleheads forum topic: "Questions on Value Averaging" (https://www.bogleheads.org/forum/viewtopic.php?t=85583). November 11, 2011

Bibliography

- Paul S. Marshall, "A Statistical Comparison Of Value Averaging vs. Dollar Cost Averaging and Random Investment Techniques (https://web.archive.org/web/20240419044005/https://www.financialdecisionsonline.org/archive/pdffiles/v13n1/marshall.pdf)", Journal of Financial and Strategic Decisions Volume 13, Number 1, Spring 2000.

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